# INVERTIS UNIVERSITY, BAREILLY

**FACULTY OF ENGINEERING & TECHNOLOGY**

OFFICIAL ACADEMIC LEARNING RESOURCE â¢ DEPARTMENT OF CSE

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**Course: [BCS-301] Data Structures**

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: PPT SLIDES**

Module / Unit: Full Syllabus **Faculty Author: Department of Computer Scie**

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## Complete Course Presentation Deck: Data Structures & Algorithms

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### Slide Module 1: Foundational Paradigms & Linear Collections

- Memory management in C: stack vs heap, pointer arithmetic, contiguous allocation.
- Abstract data type specification and interface design in engineering applications.
- Array based representations of vectors, sparse matrices, and dynamic buffers.

### Slide Module 2: Non-Linear Hierarchies & Trees

- Recursive tree traversal algorithms and structural proofs via mathematical induction.
- Huffman coding trees for lossless data compression and prefix codes.
- Multiway search trees: 2-3 trees, B-Trees, and database disk-block indexing.

### Slide Module 3: Advanced Graphs, Greedy & Dynamic Methods

- Minimum Spanning Trees: Kruskal’s algorithm (Disjoint Set Union) and Prim’s algorithm.
- Single Source Shortest Path: Dijkstra’s algorithm using Fibonacci / Binary Heaps.
- Summary: Real-world case studies in networking routing and compilation parse trees.